

ATHARVA PANSARE

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EDUCATION

North Carolina State University

Master of Science in Computer Science

Coursework: Algorithms, Neural Networks, Deep Learning, Data Science, Software Engineering, DBMS, Object-Oriented Design

May 2025

GPA: 4.0

MIT - World Peace University, India

Bachelor of Technology in Computer Science and Engineering

July 2023

CGPA: 9.78

EXPERIENCE

Data Scientist

North Carolina State University

Dec. 2024 – Present

- Optimized parameters differential equations from a published research paper to model methane gas respiration in Python (SciPy, NumPy), optimizing parameters against Mean Squared Error and statistical analysis in R
- Led data preparation for the PSI Hackathon 2025 by engineering a Python (Pandas) pipeline to ingest, clean, and integrate diverse, large-scale real world datasets including PRISM, USDA NASS, and gSSURGO

Machine Learning Engineer Intern

KPOINT Technologies Pvt. Ltd.

Jul. 2022 – Jan. 2023

- Designed and implemented an **end-to-end ML pipeline**, including preprocessing, model training, and real-time inference; built a REST API with CherryPy and containerized the system using **Docker** for seamless model deployment
- Developed a real time Thumbnail Recommendation System for the KPOINT Video Platform, using **TensorFlow** and **Transfer Learning** in Python, which had an accuracy of over 95%
- Scraped and preprocessed 1100+ thumbnails, converting them to grayscale for consistent model training and used various **data augmentation** techniques to increase the dataset size by 25%
- Automated the storage/retrieval of the dataset in **AWS S3** for easy access and scalability during model training

PROJECTS

Large Language Models, RAG - Hack NCState (2nd Place Winner) | FastAPI, React, MongoDB, Nginx

- Developed an end-to-end fraud detection web app using **Retrieval-Augmented Generation** with a **locally hosted Llama3 LLM** for real-time email and message analysis
- Configured an Nginx reverse proxy to **enforce rate limiting, handle CORS policies, and load balancing**
- Reduced response time by 5x** by migrating from local cosine similarity computation to MongoDB Atlas's vector search, **optimizing top-k retrieval** over 1,200+ message embeddings
- Built a **CI/CD pipeline** using GitHub Actions to automate builds, deployment workflows, and API testing with **pytest**

Object Detection using Deep Learning – NC PSI Hackathon (Winner) | YOLOv8, Python

- Developed an object detection model using YOLOv8 to classify 47 plant types from synthetic and real-world images
- Generated a synthetic dataset with plant cutouts; optimized preprocessing techniques to enhance model accuracy
- Utilized **AWS SageMaker** for model training and deployment; collaborated in a team of 4 over a 3-day competition

Event Management System | Flask, React, SQLite, RBAC

- Built a full-stack app with **Flask** and **React**, featuring **CRUD** operations, authentication, and role-based access

Real-time Chat Application | Node.js, React, Apache Kafka

- Built a Chat app with **WebSocket** communication via a **Node.js** server, with **MySQL** and **Kafka** for message persistence

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++, C, Java, R, HTML

Frameworks: TensorFlow, Keras, Pytorch, Flask, React, Node.js Scikit-learn, NumPy, Pandas, FastAPI, Ruby On Rails

MLOps and DevOps: Docker, Apache Kafka, Git, GitHub Actions, AWS S3, EC2, SageMaker, Nginx, Postman

Databases: MySQL, MongoDB, SQLite, PostgreSQL, SQL

RESEARCH PUBLICATIONS

First Author, Drone Detection using YOLO and SSD: A Comparative Study ([IEEE Xplore link](#))

First Author, Personality Prediction with Natural Language Processing using Questionnaire Responses ([IEEE Xplore link](#))

Author, Remote Landslide Detection Using Semantic Segmentation ([IEEE Xplore link](#))